

Day 06: Graphical Representation of Data

Understanding Data Through Visualization

100 Days of Data Science

Why Visualization is Necessary

Humans understand pictures faster than numbers. Large datasets in table form are difficult to analyze.

Graphs help to:

- Identify trends quickly
- Compare values easily
- Spot unusual patterns
- Communicate insights clearly

In Data Science, visualization is the first step of understanding data.

Real-Life Example

A company records daily sales for one year. Looking at 365 numbers does not reveal patterns easily.

But a line graph instantly shows:

- Seasonal growth
- Sudden drops
- Long-term trends

This is why businesses rely heavily on visual data.

Main Types of Graphs

Commonly used graphs include:

- Bar Chart
- Histogram
- Pie Chart
- Line Graph

Each serves a specific purpose.

Bar Chart

Used mainly for categorical data.

Real-Life Example

A school compares number of students in each class.

Bars represent:

- Class 10 – 40 students
- Class 11 – 35 students
- Class 12 – 50 students

The tallest bar shows the class with most students.

Why It Works

- Easy comparison
- Clear differences

Histogram

Used for continuous numerical data.

Real-Life Example

A hospital records patient ages.

A histogram shows:

- Most patients between 20–40
- Few patients above 70

This helps plan medical resources.

Why It Works

- Shows data distribution
- Highlights skewness
- Detects outliers

Pie Chart

Shows proportion of parts in a whole.

Real-Life Example

Company expense distribution:

- Rent – 40%
- Salary – 35%
- Marketing – 15%
- Other – 10%

The largest slice shows where most money goes.

Line Graph

Shows change over time.

Real-Life Example

Monthly temperature of a city. It shows:

- Summer peaks
- Winter drops

Used in weather forecasting and trend analysis.

Choosing the Correct Graph

- Categories → Bar chart
- Continuous values → Histogram
- Percentages → Pie chart
- Time series → Line graph

Wrong choice can mislead conclusions.

Visualization in Data Science Workflow

- Understand data before cleaning
- Identify missing or incorrect values
- Choose correct ML models
- Present insights to stakeholders

Common Visualization Mistakes

- Using pie charts for too many categories
- Ignoring scale differences
- Overcrowding graphs

Final Takeaway

Visualization converts numbers into understanding.

Good Data Science always begins with good visualization.